AIR QUALITY INDEX (AQI) ESTIMATION USING MACHINE LEARNING

Exploratory Data Analysis & Visualization — Case Study

Project Area	Exploratory Data Analysis, Visualization and Regression
Dataset	India City-Day Air Quality Dataset
Target Variable	Air Quality Index (AQI)
Models Compared	Linear Regression, Random Forest, Gradient Boosting, XGBoost
Tools	Python, Pandas, NumPy, Matplotlib, Scikit-learn, XGBoost

Student Name: Rakesh Rauniyar
Department: Computer Science and Engineering
Course Code: U21ADP05

Academic Project Report

ABSTRACT

Air pollution is a major environmental concern, and the Air Quality Index (AQI) provides a consolidated measure of ambient air quality. This project applies exploratory data analysis, data preprocessing, visualization and supervised machine-learning regression to estimate AQI from pollutant measurements, city information and calendar features. Four regression algorithms—Linear Regression, Random Forest, Gradient Boosting and XGBoost—are trained and evaluated using a chronological train-test split. Performance is measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). In the recorded experiment, Gradient Boosting achieved the lowest RMSE (31.4081) and highest R^2 (0.9087), while XGBoost achieved the lowest MAE (17.7473). The analysis also indicates that errors are larger for extreme AQI observations. The project is an AQI estimation case study and does not claim to forecast future AQI.

1. INTRODUCTION

Air quality is influenced by several pollutants, including particulate matter (PM2.5 and PM10), nitrogen oxides, ammonia, carbon monoxide, sulphur dioxide and ozone. AQI expresses the combined impact of measured pollutants in a form that is easier to interpret. Data analysis can reveal pollutant patterns, missing-value behaviour and relationships between observations and AQI. Machine learning can then learn a mapping from available pollutant measurements to an AQI estimate.

1.1 Objectives

- Explore the structure, completeness and distribution of the air-quality dataset.
- Clean the data and prepare suitable numerical and categorical input features.
- Compare four regression models using consistent train-test data.
- Visualize model performance, actual-versus-predicted values and residual behaviour.
- Identify limitations, particularly prediction errors for extreme AQI values.

2. DATASET DESCRIPTION

The project uses the India City-Day Air Quality dataset, containing daily records for multiple Indian cities. The original CSV contains 29,531 rows and 16 columns, including City, Date, pollutant concentrations, AQI and AQI_Bucket. The AQI column is the prediction target. AQI_Bucket is derived from AQI and is therefore excluded from the input features to avoid target leakage. Xylene is removed because 61.32% of its values are missing. Rows without an AQI value and rows with invalid dates are removed.

2.1 Missing data and cleaning

The initial exploration found no duplicate rows. Missingness was substantial in some fields: Xylene (61.32%), PM10 (37.72%) and NH3 (34.97%). After the cleaning operations, the dataset contained 24,850 records and 14 columns. Missing values in the remaining predictor columns were retained until preprocessing, where imputation was fitted using training data only.

2.2 Features and target

The input variables comprise City; PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene and Toluene; and calendar components Year, Month and Day extracted from Date. AQI is used as the continuous target variable.

3. METHODOLOGY

3.1 Exploratory data analysis and preprocessing

The dataset was inspected for shape, data types, missing values and duplicate records. Date values were converted to datetime format and observations were sorted chronologically. The data was divided by unique dates: dates before the split date were assigned to training and dates on or after it to testing. This produced 16,018 training records and 8,832 testing records. The training period was 1 January 2015 to 26 May 2019, and the test period was 27 May 2019 to 1 July 2020.

Numerical columns were imputed using the median of the training data. City was imputed using the most frequent training value and converted using one-hot encoding with unknown categories ignored. The preprocessing transformer was fitted only on training data and then applied to the test data. The processed matrices contained 33 features and no remaining missing values.

3.2 Machine-learning algorithms

Linear Regression: Used as a baseline model to estimate a linear relationship between input features and AQI.
Random Forest Regressor: An ensemble of decision trees. The experiment used 100 trees, maximum depth 15 and random state 42.
Gradient Boosting Regressor: Builds an additive ensemble sequentially to reduce prediction error. The experiment used 100 estimators, learning rate 0.1, maximum depth 3 and random state 42.
XGBoost Regressor: A gradient-boosted tree implementation configured with 100 estimators, learning rate 0.1, maximum depth 6 and squared-error objective.

3.3 Evaluation metrics

MAE measures the average absolute difference between actual and predicted AQI. **RMSE** penalizes larger errors more strongly because errors are squared before averaging. **R^2** indicates the proportion of target variance explained by the model relative to a mean baseline. Lower MAE and RMSE are preferred; higher R^2 is preferred.

4. RESULTS AND DISCUSSION

Model	MAE	RMSE	R^2
Gradient Boosting	18.4283	**31.4081**	**0.9087**
Random Forest	17.7693	32.3495	0.9032
XGBoost	**17.7473**	35.0770	0.8862
Linear Regression	24.9152	37.3474	0.8709

Gradient Boosting is the best model when selecting by RMSE and R^2. XGBoost has a marginally lower MAE, but its RMSE is higher, indicating that its larger errors have a stronger effect on squared-error performance. Random Forest also performs competitively. Linear Regression has the highest errors and lowest R^2 among the evaluated models.

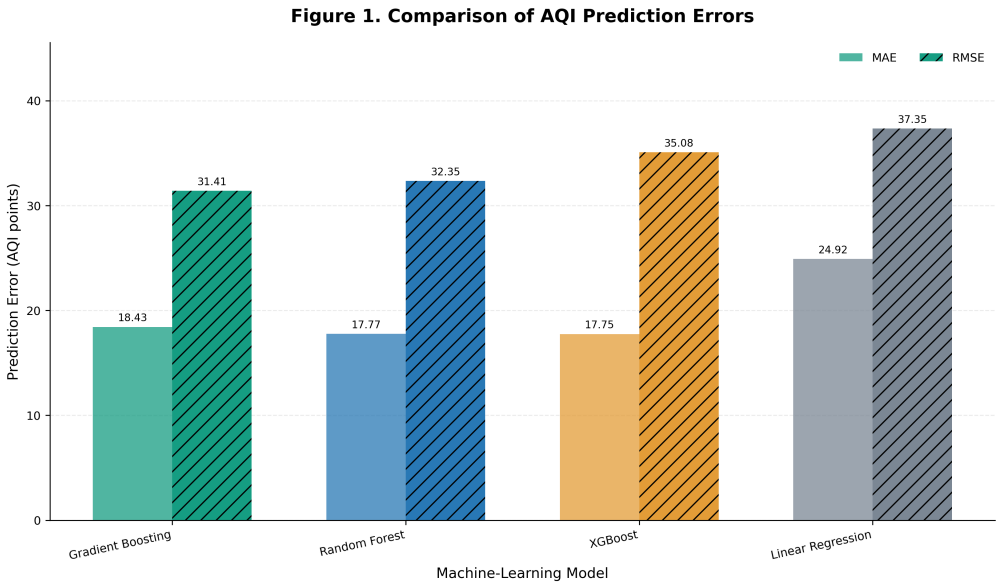


Figure 1. Error comparison of the evaluated regression models.

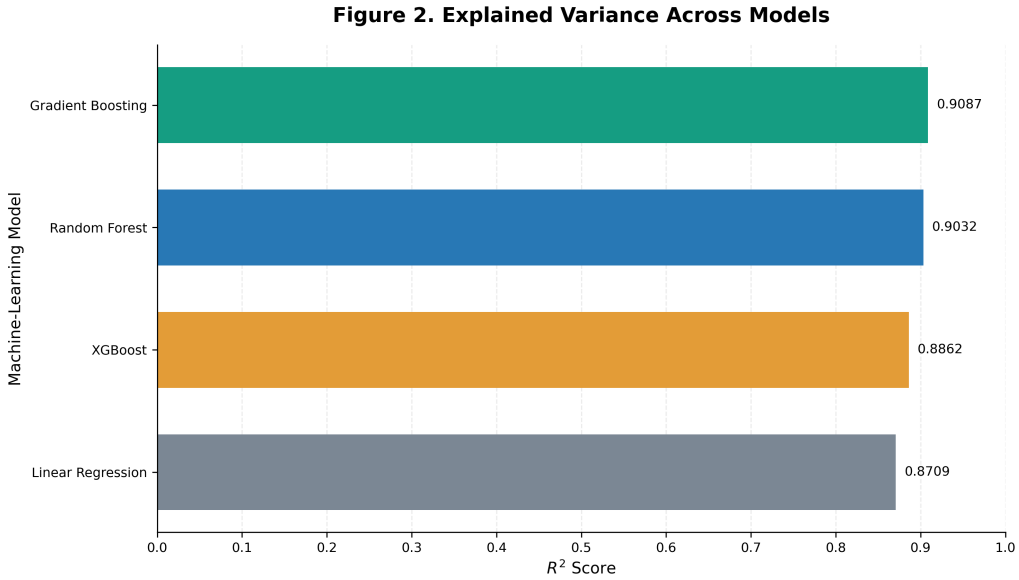


Figure 2. R² score comparison.

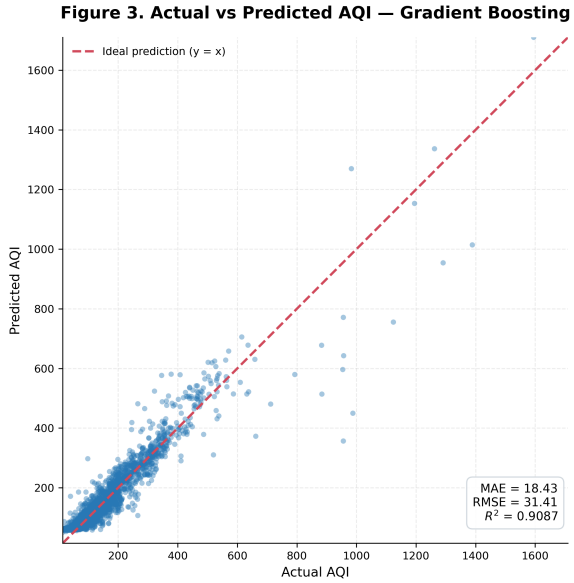


Figure 3. Actual versus predicted AQI values.

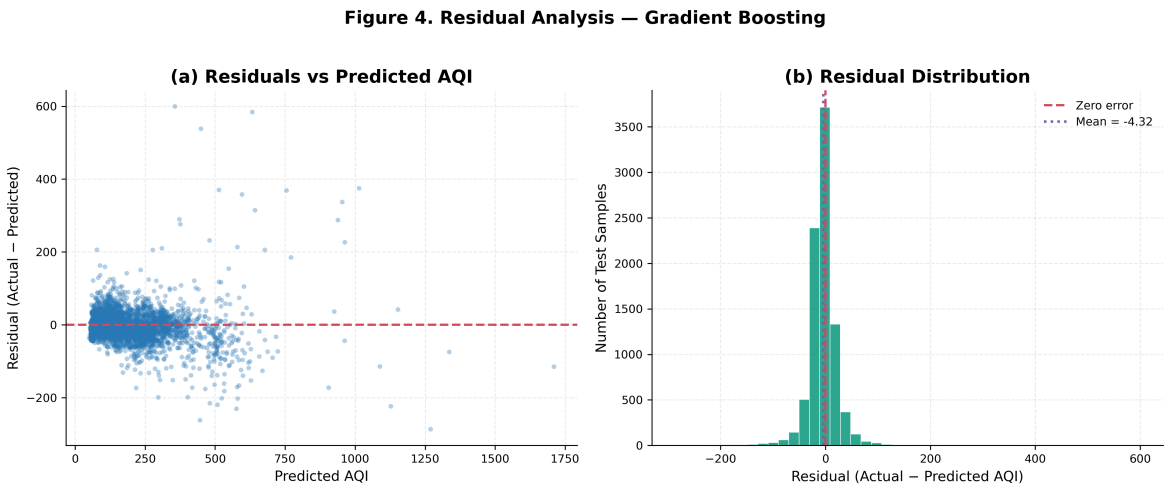


Figure 4. Residual analysis for the selected model.

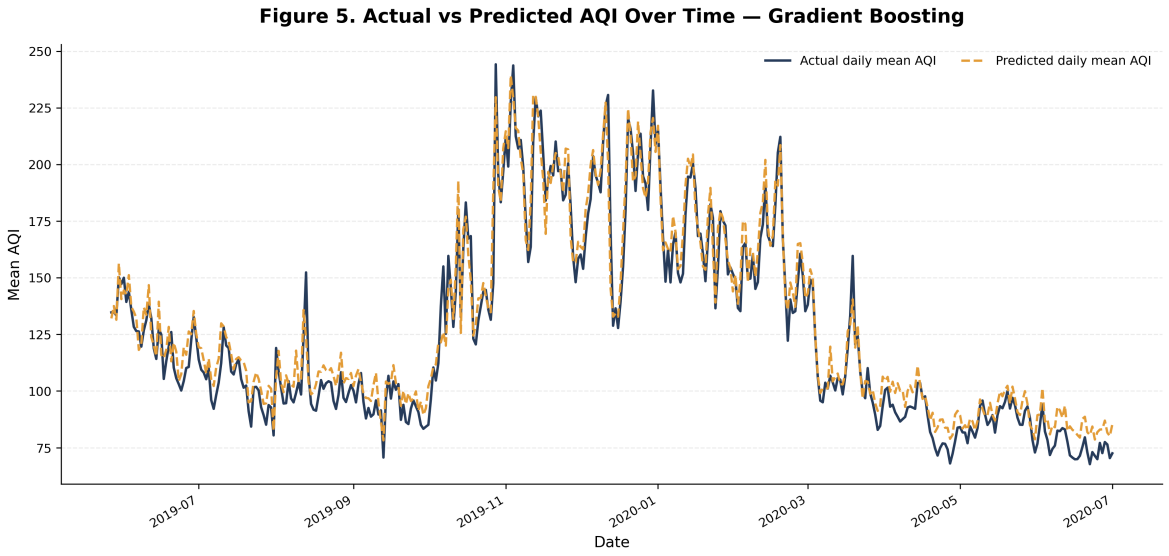


Figure 5. AQI time-series visualization from the analysis.

5. ERROR ANALYSIS

The range-wise evaluation in the analysis notebook shows that prediction performance varies across AQI ranges. For observations with actual AQI above 500, the test set contains 92 samples and Gradient Boosting has an MAE of 114.17 and RMSE of 172.33. This indicates that the model has substantially larger errors for extreme AQI observations than for many moderate-AQI observations. Such cases are relatively sparse and can be difficult for a model trained to minimize overall error.

The residual analysis compares actual and predicted values through the difference (actual AQI – predicted AQI). A residual near zero indicates a close estimate; large positive or negative residuals indicate underestimation or overestimation, respectively. Actual-versus-predicted plots help identify deviations from the ideal diagonal relationship.

6. CONCLUSION

This case study demonstrates an end-to-end workflow for AQI estimation using exploratory analysis, preprocessing, visualization and machine-learning regression. Four models were evaluated on the same chronological test set. Gradient Boosting provided the strongest overall result by RMSE (31.4081) and R^2 (0.9087), while XGBoost achieved the lowest MAE (17.7473). The results show that ensemble tree-based methods can estimate AQI effectively on this dataset, but errors increase for extreme AQI values.

7. LIMITATIONS AND FUTURE SCOPE

The dataset contains substantial missingness, and imputation may not fully represent real pollutant conditions. Extreme AQI values are relatively uncommon and are harder to estimate accurately. The chronological split is based on shared dates across cities; it is not a strict unseen-city evaluation. Future work can examine city-wise validation, more systematic treatment of extreme observations, feature-importance analysis, hyperparameter tuning and additional datasets. This project estimates AQI from available observations and does not establish future AQI forecasting capability.

REFERENCES

[1] R. Rao, “Air Quality Data in India,” Kaggle dataset. Available: https://www.kaggle.com/datasets/rohanrao/air-quality-data-in-india (accessed for this academic case study).

[2] Scikit-learn developers, “Scikit-learn: Machine Learning in Python,” https://scikit-learn.org/.

[3] XGBoost developers, “XGBoost Documentation,” https://xgboost.readthedocs.io/.

APPENDIX: REPOSITORY CONTENTS

The accompanying GitHub project package contains the analysis notebook, dataset, Python training script, requirements file, README, generated figures and experiment result files. The README provides setup and execution instructions. Before final submission, confirm the public GitHub repository URL and verify that the report PDF is below the form’s 10 MB upload limit.